

10	C	Assume the sphere is dropped from a height h. Considering its downward motion to the plate, Gain in KE = Loss in GPE, $\frac{1}{2}mu^2 = mgh$ $u^2 = 2gh$ Upon rebound, considering its upward motion to its maximum height, Loss in KE = Gain in GPE, $\frac{1}{2}mv^2 = mg\left(\frac{1}{2}h\right)$ $v^2 = gh$ Hence, $\frac{v^2}{u^2} = \frac{gh}{2gh}$ $\left(\frac{v}{u}\right)^2 = \frac{1}{2}$ $\frac{v}{u} = \frac{1}{\sqrt{2}}$
11	B	Using conservation of energy